

Course: Math 2414 Calculus II

Trigonometry Formulas

Derivative Formulas

Integration Formulas

Angles

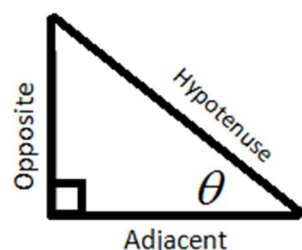
One grad equals $\frac{9}{10}$ of a degree or $\frac{\pi}{200}$ of a radian

1 full circle = 360 degrees = 2π radians = 400 grads.

Degrees	Radians	Gradients	Sin(x)	Cos(x)	Tan(x)
30°	$\frac{\pi}{6}$	33½ grad	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$
45°	$\frac{\pi}{4}$	50 grad	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\pi}{3}$	66⅔ grad	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	$\frac{\pi}{2}$	100 grad	1	0	Undefined
120°	$\frac{2\pi}{3}$	133½ grad	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$-\sqrt{3}$
135°	$\frac{3\pi}{4}$	150 grad	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	-1
150°	$\frac{5\pi}{6}$	166⅔ grad	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{\sqrt{3}} = -\frac{\sqrt{3}}{3}$
180°	π	200 grad	0	-1	0
210°	$\frac{7\pi}{6}$	233½ grad	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$
225°	$\frac{5\pi}{4}$	250 grad	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	-1
240°	$\frac{4\pi}{3}$	266⅔ grad	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$\sqrt{3}$
270°	$\frac{3\pi}{2}$	300 grad	-1	0	Undefined
300°	$\frac{5\pi}{3}$	333½ grad	$-\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$-\sqrt{3}$
315°	$\frac{7\pi}{4}$	350 grad	$-\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	-1
330°	$\frac{11\pi}{6}$	366⅔ grad	$-\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{\sqrt{3}} = -\frac{\sqrt{3}}{3}$
360°	2π	400 grad	0	1	0

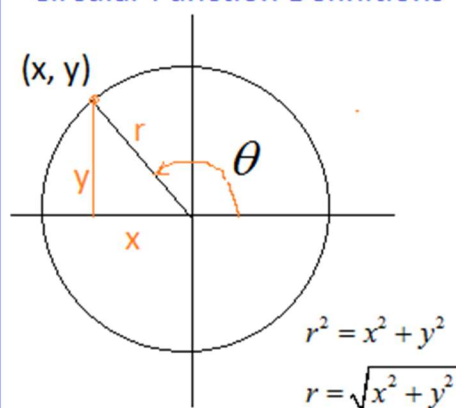
Trigonometry Formulas

Right Triange Definitions



$$\begin{aligned}\sin \theta &= \frac{\text{opp}}{\text{hyp}} & \csc \theta &= \frac{\text{hyp}}{\text{opp}} \\ \cos \theta &= \frac{\text{adj}}{\text{hyp}} & \sec \theta &= \frac{\text{hyp}}{\text{adj}} \\ \tan \theta &= \frac{\text{opp}}{\text{adj}} & \cot \theta &= \frac{\text{adj}}{\text{opp}}\end{aligned}$$

Circular Function Definitions



$$\begin{aligned}\sin \theta &= \frac{y}{r} & \csc \theta &= \frac{r}{y} \\ \cos \theta &= \frac{x}{r} & \sec \theta &= \frac{r}{x} \\ \tan \theta &= \frac{y}{x} & \cot \theta &= \frac{x}{y}\end{aligned}$$

Reciprocal Identities

$$\begin{aligned}\sin x &= \frac{1}{\csc x} & \cos x &= \frac{1}{\sec x} & \tan x &= \frac{1}{\cot x} \\ \csc x &= \frac{1}{\sin x} & \sec x &= \frac{1}{\cos x} & \cot x &= \frac{1}{\tan x}\end{aligned}$$

Quotient Identities

$$\tan x = \frac{\sin x}{\cos x} \quad \cot x = \frac{\cos x}{\sin x}$$

Pythagorean Identities

$$\begin{aligned}\sin^2 x + \cos^2 x &= 1 \\ 1 + \tan^2 x &= \sec^2 x \\ 1 + \cot^2 x &= \csc^2 x\end{aligned}$$

Cofunction Identities

$$\begin{aligned}\sin\left(\frac{\pi}{2} - x\right) &= \cos x & \cos\left(\frac{\pi}{2} - x\right) &= \sin x \\ \csc\left(\frac{\pi}{2} - x\right) &= \sec x & \tan\left(\frac{\pi}{2} - x\right) &= \cot x \\ \sec\left(\frac{\pi}{2} - x\right) &= \csc x & \cot\left(\frac{\pi}{2} - x\right) &= \tan x\end{aligned}$$

Even Identities

$$\cos(-x) = \cos(x)$$

$$\sec(-x) = \sec(x)$$

Odd Identities

$$\sin(-x) = -\sin(x)$$

$$\tan(-x) = -\tan(x)$$

$$\csc(-x) = -\csc(x)$$

$$\cot(-x) = -\cot(x)$$

Sum and Difference Formulas

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \pm \tan \alpha \tan \beta}$$

Double-Angle Formulas

$$\sin 2\theta = 2 \sin \theta \cos \theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\cos 2\theta = 2 \cos^2 \theta - 1$$

$$\cos 2\theta = 1 - 2 \sin^2 \theta$$

$$\cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$\cot 2\theta = \frac{\cot^2 \theta - 1}{2 \cot \theta}$$

Triple-Angle Formula

$$\sin 3\theta = -\sin^3 \theta + 3 \cos^2 \theta \sin \theta$$

$$\sin 3\theta = -4 \sin^3 \theta + 3 \sin \theta$$

$$\cos 3\theta = \cos^3 \theta - 3 \sin^2 \theta \cos \theta$$

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$$

$$\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$$

$$\cot 3\theta = \frac{3 \cot \theta - \cot^3 \theta}{1 - 3 \cot^2 \theta}$$

Power-Reducing Formulas

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

Sum-to-Product Formulas

$$\sin u + \sin v = 2 \sin\left(\frac{u+v}{2}\right) \cos\left(\frac{u-v}{2}\right)$$

$$\sin u - \sin v = 2 \cos\left(\frac{u+v}{2}\right) \sin\left(\frac{u-v}{2}\right)$$

$$\cos u + \cos v = 2 \cos\left(\frac{u+v}{2}\right) \cos\left(\frac{u-v}{2}\right)$$

$$\cos u - \cos v = -2 \sin\left(\frac{u+v}{2}\right) \sin\left(\frac{u-v}{2}\right)$$

Product-to-Sum Formulas

$$\sin u \sin v = \frac{1}{2} [\cos(u-v) - \cos(u+v)]$$

$$\cos u \cos v = \frac{1}{2} [\cos(u-v) + \cos(u+v)]$$

$$\sin u \cos v = \frac{1}{2} [\sin(u+v) + \sin(u-v)]$$

$$\cos u \sin v = \frac{1}{2} [\sin(u+v) - \sin(u-v)]$$

Reciprocal Identities

$$\sin x = \frac{1}{\csc x} \quad \cos x = \frac{1}{\sec x}$$

$$\csc x = \frac{1}{\sin x} \quad \sec x = \frac{1}{\cos x}$$

$$\tan x = \frac{1}{\cot x}$$

$$\cot x = \frac{1}{\tan x}$$

Quotient Identities

$$\tan x = \frac{\sin x}{\cos x} \quad \cot x = \frac{\cos x}{\sin x}$$

Pythagorean Identities

$$\sin^2 x + \cos^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

Cofunction Identities

$$\sin\left(\frac{\pi}{2} - x\right) = \cos x \quad \cos\left(\frac{\pi}{2} - x\right) = \sin x$$

$$\csc\left(\frac{\pi}{2} - x\right) = \sec x \quad \tan\left(\frac{\pi}{2} - x\right) = \cot x$$

$$\sec\left(\frac{\pi}{2} - x\right) = \csc x \quad \cot\left(\frac{\pi}{2} - x\right) = \tan x$$

Even Identities

$$\cos(-x) = \cos(x)$$

$$\sec(-x) = \sec(x)$$

Odd Identities

$$\sin(-x) = -\sin(x)$$

$$\csc(-x) = -\csc(x)$$

Sum and Difference Formulas

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \pm \tan \alpha \tan \beta}$$

Triple-Angle Formula

$$\sin 3\theta = -\sin^3 \theta + 3\cos^2 \theta \sin \theta$$

$$\sin 3\theta = -4\sin^3 \theta + 3\sin \theta$$

$$\cos 3\theta = \cos^3 \theta - 3\sin^2 \theta \cos \theta$$

$$\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$$

$$\tan 3\theta = \frac{3\tan \theta - \tan^3 \theta}{1 - 3\tan^2 \theta}$$

$$\cot 3\theta = \frac{3\cot \theta - \cot^3 \theta}{1 - 3\cot^2 \theta}$$

Sum-to-Product Formulas

$$\sin u + \sin v = 2\sin\left(\frac{u+v}{2}\right)\cos\left(\frac{u-v}{2}\right)$$

$$\sin u - \sin v = 2\cos\left(\frac{u+v}{2}\right)\sin\left(\frac{u-v}{2}\right)$$

$$\cos u + \cos v = 2\cos\left(\frac{u+v}{2}\right)\cos\left(\frac{u-v}{2}\right)$$

$$\cos u - \cos v = -2\sin\left(\frac{u+v}{2}\right)\sin\left(\frac{u-v}{2}\right)$$

Double-Angle Formulas

$$\sin 2\theta = 2\sin \theta \cos \theta = \frac{2\tan \theta}{1 + \tan^2 \theta}$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\cos 2\theta = 2\cos^2 \theta - 1$$

$$\cos 2\theta = 1 - 2\sin^2 \theta$$

$$\cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$$

$$\tan 2\theta = \frac{2\tan \theta}{1 - \tan^2 \theta}$$

$$\cot 2\theta = \frac{\cot^2 \theta - 1}{2\cot \theta}$$

Power-Reducing Formulas

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

Product-to-Sum Formulas

$$\sin u \sin v = \frac{1}{2}[\cos(u-v) - \cos(u+v)]$$

$$\cos u \cos v = \frac{1}{2}[\cos(u-v) + \cos(u+v)]$$

$$\sin u \cos v = \frac{1}{2}[\sin(u+v) + \sin(u-v)]$$

$$\cos u \sin v = \frac{1}{2}[\sin(u+v) - \sin(u-v)]$$

Definitions of Inverse Trigonometric Functions

1) $y = \sin^{-1}(x) = \arcsin(x)$ if and only if $\sin(y) = x$

Domain: $-1 \leq x \leq 1$ Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

2) $y = \cos^{-1}(x) = \arccos(x)$ if and only if $\cos(y) = x$

Domain: $-1 \leq x \leq 1$ Range: $0 \leq y \leq \pi$

3) $y = \tan^{-1}(x) = \arctan(x)$ if and only if $\tan(y) = x$

Domain: $-\infty < x < \infty$ Range: $-\frac{\pi}{2} < y < \frac{\pi}{2}$

4) $y = \cot^{-1}(x) = \operatorname{arccot}(x)$ if and only if $\cot(y) = x$

Domain: $-\infty < x < \infty$ Range: $0 < y < \pi$

5) $y = \sec^{-1}(x) = \operatorname{arcsec}(x)$ if and only if $\sec(y) = x$

Domain: $|x| \geq 1 \Leftrightarrow x \leq -1 \text{ or } x \geq 1$ Range: $0 \leq y \leq \pi; y \neq \frac{\pi}{2}$

6) $y = \csc^{-1}(x) = \operatorname{arccsc}(x)$ if and only if $\csc(y) = x$

Domain: $|x| \geq 1 \Leftrightarrow x \leq -1 \text{ or } x \geq 1$ Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}; y \neq 0$

Derivative Formulas

Differentiation Rules

$$1. \frac{d}{dx}[cu] = cu'$$

$$2. \frac{d}{dx}[u \pm v] = u' \pm v'$$

$$3. \frac{d}{dx}[uv] = uv' + vu'$$

$$4. \frac{d}{dx}\left[\frac{u}{v}\right] = \frac{vu' + uv'}{v^2}$$

$$5. \frac{d}{dx}[v] = 0$$

$$6. \frac{d}{dx}[u^n] = nu^{n-1} \cdot u'$$

$$7. \frac{d}{dx}[x] = 1$$

$$8. \frac{d}{dx}[|u|] = \frac{u}{|u|}(u')$$

$$9. \frac{d}{dx}[\ln u] = \frac{1}{u} \cdot u'$$

$$10. \frac{d}{dx}[e^u] = e^u \cdot u'$$

$$11. \frac{d}{dx}[\log_a u] = \frac{1}{(\ln a)u} \cdot u'$$

$$12. \frac{d}{dx}[a^u] = (\ln a)a^u \cdot u'$$

$$13. \frac{d}{dx}[\sin u] = (\cos u) \cdot u'$$

$$14. \frac{d}{dx}[\cos u] = (-\sin u) \cdot u'$$

$$15. \frac{d}{dx}[\tan u] = (\sec u)^2 \cdot u'$$

$$16. \frac{d}{dx}[\cot u] = -(\csc u)^2 \cdot u'$$

$$17. \frac{d}{dx}[\sec u] = (\sec u)(\tan u) \cdot u'$$

$$18. \frac{d}{dx}[\csc u] = -(\csc u)(\cot u) \cdot u'$$

$$19. \frac{d}{dx}[\arcsin u] = \frac{1}{\sqrt{1-u^2}} \cdot u'$$

$$20. \frac{d}{dx}[\arccos u] = \frac{-1}{\sqrt{1-u^2}} \cdot u'$$

$$21. \frac{d}{dx}[\arctan u] = \frac{1}{1+u^2} \cdot u'$$

$$22. \frac{d}{dx}[\operatorname{arccot} u] = \frac{-1}{1+u^2} \cdot u'$$

$$21. \frac{d}{dx} [\arctan u] = \frac{1}{1+u^2} \cdot u'$$

$$22. \frac{d}{dx} [\operatorname{arccot} u] = \frac{-1}{1+u^2} \cdot u'$$

$$23. \frac{d}{dx} [\operatorname{arcsec} u] = \frac{1}{|u|\sqrt{u^2+1}} \cdot u'$$

$$24. \frac{d}{dx} [\operatorname{arccsc} u] = \frac{-1}{|u|\sqrt{u^2-1}} \cdot u'$$

$$25. \frac{d}{dx} [\sinh u] = (\cosh u) \cdot u'$$

$$26. \frac{d}{dx} [\cosh u] = (\sinh u) \cdot u'$$

$$27. \frac{d}{dx} [\tanh u] = (\operatorname{sech} u)^2 \cdot u'$$

$$28. \frac{d}{dx} [\coth u] = -(\operatorname{csch} u)^2 \cdot u'$$

$$29. \frac{d}{dx} [\operatorname{sech} u] = -(\operatorname{sech} u)(\tanh u) \cdot u'$$

$$30. \frac{d}{dx} [\operatorname{csch} u] = -(\operatorname{csch} u)(\coth u) \cdot u'$$

$$31. \frac{d}{dx} [\sinh^{-1} u] = \frac{1}{\sqrt{u^2+1}} \cdot u'$$

$$32. \frac{d}{dx} [\cosh^{-1} u] = \frac{1}{\sqrt{u^2-1}} \cdot u'$$

$$33. \frac{d}{dx} [\tanh^{-1} u] = \frac{1}{1-u^2} \cdot u'$$

$$34. \frac{d}{dx} [\coth^{-1} u] = \frac{1}{1-u^2} \cdot u'$$

$$35. \frac{d}{dx} [\operatorname{sech}^{-1} u] = \frac{-1}{u\sqrt{1-u^2}} \cdot u'$$

$$36. \frac{d}{dx} [\operatorname{csch}^{-1} u] = \frac{-1}{|u|\sqrt{1+u^2}} \cdot u'$$

Integration Formulas

Integration Formulas



1. $\int kf(u)du = k \int f(u)du$
2. $\int [f(u) \pm g(u)] du = \int f(u)du \pm \int g(u)du$
3. $\int 1 \cdot du = u + C$
4. $\int u^n du = \frac{u^{n+1}}{n+1} + C, n \neq -1$
5. $\int \frac{1}{u} du = \ln |u| + C$
6. $\int e^u du = e^u + C$
7. $\int a^u du = \left(\frac{1}{\ln a} \right) a^u + C$
8. $\int \sin u du = -\cos u + C$
9. $\int \cos u du = \sin u + C$
10. $\int \tan u du = -\ln |\cos u| + C$
11. $\int \cot u du = \ln |\sin u| + C$
12. $\int \sec u du = \ln |\sec u + \tan u| + C$
13. $\int \csc u du = -\ln |\csc u + \cot u| + C$
14. $\int \sec^2 u du = \tan u + C$
15. $\int \csc^2 u du = -\cot u + C$
16. $\int \sec u \cdot \tan u \cdot du = \sec u + C$
17. $\int \csc u \cdot \cot u \cdot du = -\csc u + C$
18. $\int \frac{1}{\sqrt{a^2 - u^2}} du = \arcsin \left(\frac{u}{a} \right) + C$
19. $\int \frac{1}{a^2 + u^2} du = \frac{1}{a} \arctan \left(\frac{u}{a} \right) + C$
20. $\int \frac{1}{u\sqrt{u^2 - a^2}} du = \frac{1}{a} \operatorname{arcsec} \left(\frac{|u|}{a} \right) + C$

Hyperbolic Functions

- 21) $\int \cosh u du = \sinh u + C$
- 22) $\int \sinh u du = \cosh u + C$
- 23) $\int \operatorname{sech}^2 u du = \tanh u + C$
- 24) $\int \operatorname{csch}^2 u du = -\coth u + C$
- 25) $\int \operatorname{sech} u \tanh u du = -\operatorname{sech} u + C$
- 26) $\int \csc hu \coth u du = -\csc hu + C$
- 27) $\int \frac{u}{a+bu} du = \frac{1}{b^2} [bu - a \ln |a+bu|] + C$
- 28) $\int \frac{u^2}{a+bu} du = \frac{1}{b^3} \left[\frac{1}{2}(a+bu)^2 - 2a(a+bu) + a^2 \ln |a+bu| \right] + C$
- 29) $\int \frac{u}{(a+bu)^2} du = \frac{1}{b^2} \left[\frac{a}{a+bu} + \ln |a+bu| \right] + C$

$$30) \int \frac{u^2}{(a+bu)^2} du = \frac{1}{b^3} \left[bu - \frac{a^2}{a+bu} - 2a \ln|a+bu| \right] + C$$

$$31) \int \frac{u^2}{(a+bu)^3} du = \frac{1}{b^2} \left[\frac{a}{2(a+bu)^2} - \frac{1}{a+bu} \right] + C$$

$$32) \int \frac{1}{u(a+bu)} du = \frac{1}{a} \ln \left| \frac{u}{a+bu} \right| + C \quad 33) \int \frac{1}{u^2(a+bu)} du = -\frac{1}{au} + \frac{b}{a^2} \ln \left| \frac{a+bu}{u} \right| + C$$

$$34) \int \frac{1}{u(a+bu)^2} du = \frac{1}{a(a+bu)} + \frac{1}{a^2} \ln \left| \frac{u}{a+bu} \right| + C$$

$$34b) \int \frac{1}{u^2(a+bu)^2} du = \frac{1}{a^2} \left[\frac{a+2bu}{u(a+bu)} + \frac{2b}{a} \ln \left| \frac{u}{a+bu} \right| \right] + C$$

Integrals Containing $\sqrt{a+bu}$

$$35) \int u\sqrt{a+bu} du = \frac{2}{15b^2} (3bu - 2a)(a+bu)^{3/2} + C$$

$$36) \int u^2\sqrt{a+bu} du = \frac{2}{105b^3} (15b^2u^2 - 12abu + 8a^2)(a+bu)^{3/2}$$

$$37) \int u^n\sqrt{a+bu} du = \frac{2u^n(a+bu)^{3/2}}{b(2n+3)} - \frac{2an}{b(2n+3)} \int u^{n-1}\sqrt{a+bu}$$

$$38) \int \frac{u}{\sqrt{a+bu}} du = \frac{2}{3b^2} (bu - 2a)\sqrt{a+bu} + C$$

$$39) \int \frac{u^2}{\sqrt{a+bu}} du = \frac{2}{15b^2} (3b^2u^2 - 4abu + 8a^2)\sqrt{a+bu} + C$$

$$40) \int \frac{u^n}{\sqrt{a+bu}} du = \frac{2u^n\sqrt{a+bu}}{b(2n+1)} - \frac{2an}{b(2n+1)} \int \frac{u^{n-1}}{\sqrt{a+bu}} du$$

$$41) \int \frac{1}{u\sqrt{a+bu}} du = \begin{cases} \frac{1}{\sqrt{a}} \ln \left| \frac{\sqrt{a+bu} - \sqrt{a}}{\sqrt{a+bu} + \sqrt{a}} \right| + C & \text{if } a > 0 \\ \frac{2}{\sqrt{-a}} \tan^{-1} \left(\sqrt{\frac{a+bu}{-a}} \right) + C & \text{if } a < 0 \end{cases}$$

$$42) \int \frac{1}{u^n \sqrt{a+bu}} du = -\frac{\sqrt{a+bu}}{a(n-1)u^{n-1}} - \frac{b(2n-3)}{2a(n-1)} \int \frac{1}{u^{n-1} \sqrt{a+bu}} du$$

$$44) \int \frac{\sqrt{a+bu}}{u^n} du = -\frac{(a+bu)^{3/2}}{a(n-1)u^{n-1}} - \frac{b(2n-5)}{2a(n-1)} \int \frac{\sqrt{a+bu}}{u^{n-1}} du$$

Integrals Containing $a^2 \pm u^2$ ($a > 0$)

$$45) \int \frac{1}{a^2 + u^2} du = \frac{1}{a} \tan^{-1} \left(\frac{u}{a} \right) + C$$

$$46) \int \frac{1}{a^2 - u^2} du = \frac{1}{2a} [\ln|u-a| - \ln|u+a|] + C$$

$$47) \int \frac{1}{u^2 - a^2} du = -\frac{1}{2a} [\ln|u-a| - \ln|u+a|] + C$$

Integrals Containing $\sqrt{u^2 \pm a^2}$ ($a > 0$)

$$48) \int \frac{1}{\sqrt{u^2 \pm a^2}} du = \ln|u + \sqrt{u^2 \pm a^2}| + C$$

$$49) \int \sqrt{u^2 \pm a^2} du = \frac{u}{2} \sqrt{u^2 \pm a^2} \pm \frac{a^2}{2} \ln|u + \sqrt{u^2 \pm a^2}| + C$$

$$50) \int u \sqrt{u^2 \pm a^2} du = \frac{1}{3} (u^2 \pm a^2)^{3/2} + C$$

$$51) \int u^2 \sqrt{u^2 \pm a^2} du = \frac{u}{8} (2u^2 \pm a^2) \sqrt{u^2 \pm a^2} - \frac{a^4}{8} \ln|u + \sqrt{u^2 \pm a^2}| + C$$

$$52) \int \frac{\sqrt{u^2 + a^2}}{u} du = \sqrt{u^2 + a^2} - a \ln \left| \frac{a + \sqrt{u^2 + a^2}}{u} \right| + C$$

$$53) \int \frac{\sqrt{u^2 - a^2}}{u} du = \sqrt{u^2 - a^2} - a \sec^{-1} \left(\frac{u}{a} \right) + C$$

$$54) \int \frac{\sqrt{u^2 \pm a^2}}{u^2} du = -\frac{\sqrt{u^2 \pm a^2}}{u} + \ln \left| u + \sqrt{u^2 \pm a^2} \right| + C$$

$$55) \int \frac{u^2}{\sqrt{u^2 \pm a^2}} du = \frac{u}{2} \sqrt{u^2 \pm a^2} \mp \frac{a^2}{2} \ln \left| u + \sqrt{u^2 \pm a^2} \right| + C$$

$$56) \int \frac{1}{u\sqrt{u^2 + a^2}} du = -\frac{1}{a} \ln \left| \frac{a + \sqrt{u^2 + a^2}}{u} \right| + C$$

$$57) \int \frac{1}{u\sqrt{u^2 - a^2}} du = \frac{1}{a} \operatorname{arcsec} \left(\frac{|u|}{a} \right) + C$$

$$58) \int \frac{1}{u^2 \sqrt{u^2 \pm a^2}} du = \mp \frac{\sqrt{u^2 \pm a^2}}{a^2 u} + C$$

$$59) \int (u^2 \pm a^2)^{3/2} du = \frac{u}{8} (2u^2 \pm 5a^2) \sqrt{u^2 \pm a^2} + \frac{3a^4}{8} \ln \left| u + \sqrt{u^2 \pm a^2} \right| + C$$

$$60) \int \frac{1}{(u^2 \pm a^2)^{3/2}} du = \pm \frac{u}{a^2 \sqrt{u^2 \pm a^2}} + C$$

$$60b) \int \frac{1}{(a^2 + u^2)^2} du = \frac{1}{2a^3} \left[\frac{au}{a^2 + u^2} + \tan^{-1} \left(\frac{u}{a} \right) \right] + C$$

Integrals Containing $\sqrt{a^2 - u^2}$ ($a > 0$)

$$61) \int \frac{1}{\sqrt{a^2 - u^2}} du = \sin^{-1} \left(\frac{u}{a} \right) + C$$

$$62) \int \sqrt{a^2 - u^2} du = \frac{u}{2} \sqrt{a^2 - u^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{u}{a} \right) + C$$

$$63) \int u^2 \sqrt{a^2 - u^2} du = \frac{u}{8} (2u^2 - a^2) \sqrt{a^2 - u^2} + \frac{a^4}{8} \sin^{-1} \left(\frac{u}{a} \right) + C$$

$$64) \int \frac{\sqrt{a^2 - u^2}}{u} du = \sqrt{a^2 - u^2} - a \ln \left| \frac{a + \sqrt{a^2 - u^2}}{u} \right| + C$$

$$65) \int \frac{\sqrt{a^2 - u^2}}{u^2} du = -\frac{\sqrt{a^2 - u^2}}{u} - \sin^{-1} \left(\frac{u}{a} \right) + C$$

$$66) \int \frac{u^2}{\sqrt{a^2 - u^2}} du = -\frac{u}{2} \sqrt{a^2 - u^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{u}{a} \right) + C$$

$$67) \int \frac{1}{u \sqrt{a^2 - u^2}} du = -\frac{1}{a} \ln \left| \frac{a + \sqrt{a^2 - u^2}}{u} \right| + C$$

$$68) \int \frac{1}{u^2 \sqrt{a^2 - u^2}} du = -\frac{\sqrt{a^2 - u^2}}{a^2 u} + C$$

$$69) \int (a^2 - u^2)^{3/2} du = -\frac{u}{8} (2u^2 - 5a^2) \sqrt{a^2 - u^2} + \frac{3a^4}{8} \sin^{-1} \left(\frac{u}{a} \right) + C$$

$$70) \int \frac{1}{(a^2 - u^2)^{3/2}} du = \frac{u}{a^2 \sqrt{a^2 - u^2}} + C$$

Integrals Containing $2au - u^2$

$$71) \int \sqrt{2au - u^2} du = \frac{u-a}{2} \sqrt{2au - u^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{u-a}{a} \right) + C$$

$$72) \int u \sqrt{2au - u^2} du = \frac{2u^2 - au - 3a^2}{6} \sqrt{2au - u^2} + \frac{a^3}{2} \sin^{-1} \left(\frac{u-a}{a} \right) + C$$

$$73) \int \frac{\sqrt{2au - u^2}}{u} du = \sqrt{2au - u^2} + a \sin^{-1} \left(\frac{u-a}{a} \right) + C$$

$$74) \int \frac{\sqrt{2au - u^2}}{u^2} du = -\frac{2\sqrt{2au - u^2}}{u} - \sin^{-1} \left(\frac{u-a}{a} \right) + C$$

$$75) \int \frac{1}{\sqrt{2au - u^2}} du = \sin^{-1} \left(\frac{u-a}{a} \right) + C$$

$$76) \int \frac{u}{\sqrt{2au - u^2}} du = -\sqrt{2au - u^2} + a \sin^{-1} \left(\frac{u-a}{a} \right) + C$$

$$77) \int \frac{u^2}{\sqrt{2au - u^2}} du = -\frac{(u+3a)}{2} \sqrt{2au - u^2} + \frac{3a^2}{2} \sin^{-1} \left(\frac{u-a}{a} \right) + C$$

$$78) \int \frac{1}{u\sqrt{2au-u^2}} du = -\frac{\sqrt{2au-u^2}}{au} + C$$

$$79) \int \frac{1}{(2au-u^2)^{3/2}} du = \frac{u-a}{a^2\sqrt{2au-u^2}} + C$$

$$80) \int \frac{u}{(2au-u^2)^{3/2}} du = \frac{u}{a\sqrt{2au-u^2}} + C$$

Integrals Containing Trigonometric Functions

$$81) \int \sin^2 u du = \frac{1}{2}u - \frac{1}{4}\sin 2u + C$$

$$82) \int \cos^2 u du = \frac{1}{2}u + \frac{1}{4}\sin 2u + C$$

$$83) \int \tan^2 u du = \tan u - u + C$$

$$84) \int \cot^2 u du = -\cot u - u + C$$

$$85) \int \sin^n u du = -\frac{1}{n}\sin^{n-1} u \cos u + \frac{n-1}{n} \int \sin^{n-2} u du + C$$

$$86) \int \cos^n u du = -\frac{1}{n}\cos^{n-1} u \sin u + \frac{n-1}{n} \int \cos^{n-2} u du + C$$

$$87) \int \tan^n u du = \frac{1}{n-1}\tan^{n-1} u - \int \tan^{n-2} u du + C$$

$$88) \int \cot^n u du = -\frac{1}{n-1}\cot^{n-1} u - \int \cot^{n-2} u du + C$$

$$89) \int \sec^n u du = \frac{1}{n-1} \sec^{n-2} u \tan u + \frac{n-2}{n-1} \int \sec^{n-2} u du + C$$

$$90) \int \csc^n u du = -\frac{1}{n-1} \csc^{n-2} u \cot u + \frac{n-2}{n-1} \int \csc^{n-2} u du + C$$

$$91) \int \sin mu \cdot \cos nu du = -\frac{\cos(m+n)u}{2(m+n)} - \frac{\cos(m-n)u}{2(m-n)} + C$$

$$92) \int \cos mu \cdot \cos nu du = \frac{\sin(m+n)u}{2(m+n)} + \frac{\sin(m-n)u}{2(m-n)} + C + C$$

$$93) \int \sin mu \cdot \cos nu du = -\frac{\cos(m+n)u}{2(m+n)} - \frac{\cos(m-n)u}{2(m-n)} + C$$

$$94) \int u \sin u du = \sin u - u \cos u + C$$

$$95) \int u \cos u du = \cos u + u \sin u + C$$

$$96) \int u^2 \sin u du = 2u \sin u + (2 - u^2) \cos u + C$$

$$97) \int u^2 \cos u du = 2u \cos u + (u^2 - 2) \sin u + C$$

$$98) \int u^n \sin u du = -u^n \cos u + n \int u^{n-1} \cos u du + C$$

$$99) \int u^n \cos u du = u^n \sin u - n \int u^{n-1} \sin u du + C$$

$$100) \int \sin^m u \cdot \cos^n u du = -\frac{\sin^{m-1} u \cdot \cos^{n+1} u}{m+n} + \frac{m-1}{m+n} \int \sin^{m-2} u \cos^n u du$$

$$\text{For } m=1 \text{ and } n=1: \int \sin u \cos u du = \frac{(\cos u)^2}{2} + C$$

$$100b) \int \frac{1}{1 \pm \tan u} du = \frac{1}{2} (u \pm \ln |\cos u \pm \sin u|) + C$$

$$100c) \int \frac{1}{1 \pm \cot u} du = \frac{1}{2} (u \mp \ln |\sin u \pm \cos u|) + C$$

$$100d) \int \frac{1}{1 \pm \sec u} du = u + \cot u \mp \csc u + C$$

$$100e) \int \frac{1}{1 \pm \csc u} du = u - \tan u \pm \sec u + C$$

Integrals Containing Inverse Trigonometric Functions

$$101) \int \sin^{-1} u du = u \cdot \sin^{-1} u + \sqrt{1-u^2} + C$$

$$101b) \int u \sin^{-1} u du = \frac{2u^2-1}{4} \sin^{-1} u + \frac{u\sqrt{1-u^2}}{4} + C$$

$$102) \int \cos^{-1} u du = u \cdot \cos^{-1} u - \sqrt{1-u^2} + C$$

$$102b) \int u \cos^{-1} u du = \frac{2u^2-1}{4} \cos^{-1} u - \frac{u\sqrt{1-u^2}}{4} + C$$

$$103) \int \tan^{-1} u du = u \cdot \tan^{-1} u - \ln \sqrt{1+u^2} + C$$

$$103b) \int u \tan^{-1} u du = \frac{u^2+1}{2} \tan^{-1} u - \frac{u}{2} + C$$

$$104) \int \cot^{-1} u du = u \cdot \cot^{-1} u + \ln \sqrt{1+u^2} + C$$

$$105) \int \sec^{-1} u du = u \cdot \sec^{-1} u - \ln |u + \sqrt{u^2-1}| + C$$

$$106) \int \csc^{-1} u du = u \cdot \csc^{-1} u + \ln |u + \sqrt{u^2-1}| + C$$

Integrals Containing Exponential and Logarithmic Functions

$$107) \int e^{au} du = \frac{1}{a} e^{au} + C \quad \text{Note: } \int e^u du = e^u + C$$

$$108) \int a^{bu} du = \frac{1}{b} \frac{1}{\ln a} a^u + C \quad \text{Note: } \int a^u du = \frac{1}{\ln a} a^u + C$$

$$109) \int \frac{1}{au+b} du = \frac{1}{a} \cdot \ln|au+b| + C \quad \text{Note: } \int \frac{1}{u} du = \ln|u| + C;$$

$$110) \int u e^u du = e^u (u-1) + C$$

$$111) \int u^n e^u du = u^n e^u - n \int u^{n-1} e^u du + C$$

$$112) \int u^n a^u du = \frac{u^n a^u}{\ln a} - \frac{n}{\ln a} \int u^{n-1} a^u du + C$$

$$113) \int \frac{e^u}{u^n} du = -\frac{e^u}{(n-1)u^{n-1}} + \frac{1}{n-1} \int \frac{e^u}{u^{n-1}} du + C$$

$$114) \int \frac{a^u}{u^n} du = -\frac{a^u}{(n-1)u^{n-1}} + \frac{\ln a}{n-1} \int \frac{a^u}{u^{n-1}} du + C$$

$$115) \int \ln u du = u \ln u - u + C$$

$$\int (\ln u)^2 du = u \left[2 - 2 \ln u + (\ln u)^2 \right] + C$$

$$\int (\ln u)^n du = u (\ln u)^n - n \int (\ln u)^{n-1} du + C$$

$$116) \int u^n \ln u du = \frac{u^{n+1}}{(n+1)^2} \left[(n+1) \ln u - 1 \right] + C$$

$$117) \int \frac{1}{u \ln u} du = \ln|\ln u| + C$$

$$118) \int e^{au} \sin bu du = \frac{e^{au}}{a^2 + b^2} (a \sin bu - b \cos bu) + C$$

$$119) \int e^{au} \cos bu du = \frac{e^{au}}{a^2 + b^2} (a \cos bu + b \sin bu) + C$$

$$119b) \int \frac{1}{1+e^u} du = u - \ln(1+e^u) + C$$

Wallis Formulas

$$120) \int_0^{\pi/2} \sin^n u du = \int_0^{\pi/2} \cos^n u du = \frac{1 \cdot 3 \cdot 5 \cdot 7 \cdot \dots \cdot (n-2)(n-1)}{2 \cdot 4 \cdot 6 \cdot 8 \cdot \dots \cdot (n-1)(n)} \quad \text{if } n \text{ is even; } n \geq 2$$

$$121) \int_0^{\pi/2} \sin^n u du = \int_0^{\pi/2} \cos^n u du = \frac{2 \cdot 3 \cdot 6 \cdot 8 \cdot \dots \cdot (n-2)(n-1)}{3 \cdot 5 \cdot 7 \cdot 9 \cdot \dots \cdot (n-1)(n)} \quad \text{if } n \text{ is odd; } n \geq 3$$